

Analysis-0 2026 Spring

Homework 8

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*All of my homework is available at <https://github.com/lacampanella233/Homework.git>.

Problem 1 (5-1). Let f be defined for all real numbers. Assume that for every pair of real numbers x and y ,

$$|f(x) - f(y)| \leq (x - y)^2.$$

Prove that f is a constant function.

Proof. If there exists $x < y$ such that $f(x) \neq f(y)$, then take a positive integer N such that $(y - x)^2 < N |f(y) - f(x)|$. Divide the interval $[x, y]$ into N equal subintervals, and let $x_0 = x, x_1, \dots, x_N = y$ be the endpoints of these subintervals, such that $x_k - x_{k-1} = (y - x)/N := \Delta$. Then for any $1 \leq k \leq N$,

$$|f(x_k) - f(x_{k-1})| \leq (x_k - x_{k-1})^2 = \Delta^2.$$

Summing these inequalities from $k = 1$ to $k = N$, we get

$$|f(y) - f(x)| \leq \sum_{k=1}^N |f(x_k) - f(x_{k-1})| \leq N \Delta^2 = \frac{(y - x)^2}{N}.$$

This contradicts our assumption that $(y - x)^2 < N |f(y) - f(x)|$. Therefore, $f(x) = f(y)$ for all $x, y \in \mathbb{R}$, and f is a constant function. $\square$

Problem 2 (5-4). Let $C_0, \dots, C_n$ be real constants, and suppose

$$C_0 + \frac{C_1}{2} + \dots + \frac{C_{n-1}}{n} + \frac{C_n}{n+1} = 0.$$

Prove that the equation

$$C_0 + C_1 x + \dots + C_{n-1} x^{n-1} + C_n x^n = 0$$

has at least one real root in $(0, 1)$.

Proof. Define

$$f(x) = C_0 x + \frac{C_1}{2} x^2 + \dots + \frac{C_{n-1}}{n} x^n + \frac{C_n}{n+1} x^{n+1}.$$

Then $f(0) = 0$ and $f(1) = C_0 + \frac{C_1}{2} + \dots + \frac{C_{n-1}}{n} + \frac{C_n}{n+1} = 0$. By Rolle's theorem, there exists some $c \in (0, 1)$ such that $f'(c) = 0$. Note that

$$f'(x) = C_0 + C_1 x + \dots + C_{n-1} x^{n-1} + C_n x^n.$$

Therefore, $f'(c) = 0$ implies that the equation $C_0 + C_1 x + \dots + C_{n-1} x^{n-1} + C_n x^n = 0$ has at least one real root in $(0, 1)$. $\square$

Problem 3 (5-5). Let f be defined and differentiable for all $x > 0$, and assume that $f'(x) \rightarrow 0$ as $x \rightarrow +\infty$.

Define

$$g(x) = f(x+1) - f(x).$$

Prove that $g(x) \rightarrow 0$ as $x \rightarrow +\infty$.

Proof. By the mean value theorem, for each $x > 0$, there exists some $c_x \in (x, x+1)$ such that

$$g(x) = f(x+1) - f(x) = f'(c_x)(x+1-x) = f'(c_x).$$

Since $c_x \rightarrow +\infty$ as $x \rightarrow +\infty$ and $f'(x) \rightarrow 0$ as $x \rightarrow +\infty$, we have $g(x) = f'(c_x) \rightarrow 0$ as $x \rightarrow +\infty$. $\square$

Problem 4 (5-6). Assume:

(a) f is continuous on $x \geq 0$;

(b) $f'(x)$ exists for $x > 0$;

(c) $f(0) = 0$;

(d) f' is increasing.

Let

$$g(x) = \frac{f(x)}{x} \quad (x > 0).$$

Prove that g is increasing.

Proof. Suppose the contrary, that is, there exist $0 < a < b$ such that $g(a) > g(b)$. By the mean value theorem, there exists some $c \in (a, b)$ such that

$$f(b) - f(a) = f'(c)(b-a).$$

Since $g(a) > g(b)$, we have

$$f'(c) = \frac{f(b) - f(a)}{b-a} = \frac{g(b)b - g(a)a}{b-a} < \frac{g(a)b - g(a)a}{b-a} = g(a).$$

By the mean value theorem again, there exists some $d \in (0, a)$ such that

$$f(a) - f(0) = f'(d)(a-0) \implies f'(d) = g(a).$$

Since f' is increasing and $d < c$, we have $f'(d) \leq f'(c)$, which contradicts $f'(d) = g(a) > f'(c)$. Therefore, g is increasing. $\square$

Problem 5 (5-8). Assume f' is continuous on $[a, b]$, $\varepsilon > 0$, and $a \leq x \leq b$, $a \leq t \leq b$. Prove that there exists $\delta > 0$ such that whenever $0 < |t-x| < \delta$, one has

$$\left| \frac{f(t) - f(x)}{t-x} - f'(x) \right| < \varepsilon.$$

(Equivalently, if f' is continuous on $[a, b]$, then f is uniformly differentiable on $[a, b]$.) Does this remain true for vector-valued functions?

Proof. Because f' is continuous on $[a, b]$, we have f' is uniformly continuous on $[a, b]$. Therefore, for any $\varepsilon > 0$, there exists some $\delta > 0$ such that for any $t' \in [a, b]$ with $|t' - x| < \delta$, we have $|f'(t') - f'(x)| < \varepsilon$.

For any $t \in [a, b]$ with $0 < |t-x| < \delta$, by the mean value theorem, there exists some $c_t \in (x, t)$ or (t, x) such that

$$\frac{f(t) - f(x)}{t-x} = f'(c_t).$$

Obviously $|c_t - x| \leq |t - x| < \delta$, then

$$\left| \frac{f(t) - f(x)}{t - x} - f'(x) \right| = |f'(c_t) - f'(x)| < \varepsilon.$$

The statement also remains true for vector-valued functions. Just apply the conclusion to each component of the vector-valued function, and we can get the desired result. $\square$

Problem 6 (5-9). Let f be a continuous real-valued function on $\mathbb{R}$. Assume that $f'(x)$ exists for every $x \neq 0$, and that $f'(x) \rightarrow 3$ as $x \rightarrow 0$. Question: does $f'(0)$ necessarily exist?

Solution. Yes. We will prove that

$$f'(0) = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x} = 3.$$

First of all, by $\lim_{x \rightarrow 0} f'(x) = 3$, we know that f is differentiable at some neighbourhood of 0. For any $\varepsilon > 0$, there exists a $\delta > 0$ such that for all $0 < |x| < \delta$, we have $|f'(x) - 3| < \varepsilon$. Then for all $0 < |x| < \delta$, by the mean value theorem, there exists some $c_x \in (0, x)$ or $(x, 0)$ such that

$$\frac{f(x) - f(0)}{x} = f'(c_x) \in (3 - \varepsilon, 3 + \varepsilon).$$

This implies that $\lim_{x \rightarrow 0} (f(x) - f(0))/x = 3$, and thus $f'(0)$ exists and equals 3.

Problem 7 (5-10). Let f and g be complex-valued differentiable functions on $(0, 1)$. As $x \rightarrow 0$, assume

$$f(x) \rightarrow 0, \quad g(x) \rightarrow 0, \quad f'(x) \rightarrow A, \quad g'(x) \rightarrow B,$$

where A, B are complex numbers and $B \neq 0$. Prove that

$$\lim_{x \rightarrow 0} \frac{f(x)}{g(x)} = \frac{A}{B}.$$

Compare with Example 5.18. Hint:

$$\frac{f(x)}{g(x)} = \left(\frac{f(x)}{x} - A \right) \frac{x}{g(x)} + A \frac{x}{g(x)}.$$

Apply Theorem 5.13 to the real and imaginary parts of $f(x)/x$ and $g(x)/x$.

Proof. Use the same method as in Problem 5-9, we can prove that $\lim_{x \rightarrow 0} f(x)/x = A$ and $\lim_{x \rightarrow 0} g(x)/x = B$. Then

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{f(x)}{g(x)} &= \lim_{x \rightarrow 0} \left(\frac{f(x)}{x} - A \right) \frac{x}{g(x)} + \lim_{x \rightarrow 0} A \frac{x}{g(x)} \\ &= \lim_{x \rightarrow 0} \left(\frac{f(x)}{x} - A \right) \lim_{x \rightarrow 0} \frac{x}{g(x)} + A \lim_{x \rightarrow 0} \frac{x}{g(x)} \\ &= 0 \cdot \frac{1}{B} + A \cdot \frac{1}{B} = \frac{A}{B}. \end{aligned}$$

$\square$

Problem 8 (5-11). Let f be defined in a neighborhood of x , and assume $f''(x)$ exists. Prove that

$$\lim_{h \rightarrow 0} \frac{f(x+h) + f(x-h) - 2f(x)}{h^2} = f''(x).$$

Give an example showing that this limit may still exist even when $f''(x)$ does not exist.

Proof. Since $f''(x)$ exists, by the Taylor's formula with the Peano's form of the remainder, we have

$$f(x+h) = f(x) + f'(x)h + \frac{f''(x)}{2}h^2 + o(h^2).$$

Then

$$\begin{aligned} & \lim_{h \rightarrow 0} \frac{f(x+h) + f(x-h) - 2f(x)}{h^2} \\ &= \lim_{h \rightarrow 0} \frac{1}{h^2} \left[\left(f(x) + f'(x)h + \frac{f''(x)}{2}h^2 + o(h^2) \right) + \left(f(x) - f'(x)h + \frac{f''(x)}{2}h^2 + o(h^2) \right) - 2f(x) \right] \\ &= \lim_{h \rightarrow 0} \frac{f''(x)h^2 + o(h^2)}{h^2} \\ &= f''(x). \end{aligned}$$

□

Problem 9 (5-14). Let f be a differentiable real function on (a, b) . Prove that f is convex if and only if f' is increasing. Assume further that $f''(x)$ exists for all $x \in (a, b)$. Prove that f is convex if and only if $f''(x) \geq 0$ for all $x \in (a, b)$.

Proof. “ $\implies$ ”: Suppose f is convex, we will prove that for any $x < y$, we have $f'(x) \leq f'(y)$.

For any $\xi \in N_{(y-x)/3}(x)$ and $\eta \in N_{(y-x)/3}(y)$, since $\max\{\xi, x\} < \min\{b, y\}$ and f is convex, we have

$$\frac{f(\xi) - f(x)}{\xi - x} \leq \frac{f(\eta) - f(y)}{\eta - y}.$$

Letting $\xi \rightarrow x$ and $\eta \rightarrow y$, we get $f'(x) \leq f'(y)$.

“ $\impliedby$ ”: Suppose f' is increasing, we will prove that for any $x < y$ and $\lambda \in (0, 1)$, we have

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y).$$

By the mean value theorem, there exists some $\xi \in (x, \lambda x + (1 - \lambda)y)$ and $\eta \in (\lambda x + (1 - \lambda)y, y)$ such that

$$\frac{f(\lambda x + (1 - \lambda)y) - f(x)}{\lambda x + (1 - \lambda)y - x} = f'(\xi), \quad \frac{f(y) - f(\lambda x + (1 - \lambda)y)}{y - (\lambda x + (1 - \lambda)y)} = f'(\eta).$$

Note that $\xi \leq \eta$, then $f'(\xi) \leq f'(\eta)$, which implies that

$$\frac{f(\lambda x + (1 - \lambda)y) - f(x)}{\lambda x + (1 - \lambda)y - x} \leq \frac{f(y) - f(\lambda x + (1 - \lambda)y)}{y - (\lambda x + (1 - \lambda)y)}.$$

This is equivalent to $f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$.

For the second part, note that

$$f''(x) \geq 0, \forall x \in (a, b) \Leftrightarrow f' \text{ is increasing on } (a, b).$$

Hence the conclusion follows from the first part. □

Problem 10 (5-17). Let f be a real-valued function on $[-1, 1]$ that is three times differentiable, and suppose

$$f(-1) = 0, \quad f(0) = 0, \quad f(1) = 1, \quad f'(0) = 0.$$

Prove that there exists some $x \in (-1, 1)$ such that

$$f^{(3)}(x) \geq 3.$$

Note: for the function $\frac{1}{2}(x^3 + x^2)$, equality is attained.

Proof. Denote $g(x) = f(x) - \frac{1}{2}(x^3 + x^2)$, then $g(-1) = g(0) = g(1) = g'(0) = 0$, and $g^{(3)}(x) = f^{(3)}(x) - 3$.

Suppose the contrary, that is, $g^{(3)}(x) < 0$ for all $x \in (-1, 1)$, then g'' is strictly decreasing on $(-1, 1)$. Then, either $g''(x) > 0$ for all $x \in (0, 1)$ or $g''(x) < 0$ for all $x \in (-1, 0)$.

If $g''(x) > 0$ for all $x \in (0, 1)$, then g' is strictly increasing on $(0, 1)$. Since $g'(0) = 0$, we have $g'(x) > 0$ for all $x \in (0, 1)$. This contradicts the fact that $g(1) = 0$.

If $g''(x) < 0$ for all $x \in (-1, 0)$, then we can get a contradiction in a similar way. Therefore, there exists some $x \in (-1, 1)$ such that $g^{(3)}(x) \geq 0$, which implies that $f^{(3)}(x) \geq 3$. $\square$

Problem 11 (5-19). Let f be defined on $(-1, 1)$ and assume $f'(0)$ exists. Let $-1 < a_n < b_n < 1$ with $a_n \rightarrow 0$ and $b_n \rightarrow 0$ as $n \rightarrow \infty$. Define the difference quotient

$$D_n = \frac{f(b_n) - f(a_n)}{b_n - a_n}.$$

Prove the following statements:

- (a) If $a_n < 0 < b_n$, then $\lim D_n = f'(0)$.
- (b) If $0 < a_n < b_n$ and $\frac{b_n}{b_n - a_n}$ is bounded, then $\lim D_n = f'(0)$.
- (c) If f' is continuous on $(-1, 1)$, then $\lim D_n = f'(0)$.

Give an example where f is differentiable on $(-1, 1)$ but f' is discontinuous at 0, and $a_n, b_n \rightarrow 0$ such that $\lim D_n$ exists but $\lim D_n \neq f'(0)$.

Proof. (a) WLOG, assume $f'(0) > 0$. For any $\varepsilon > 0$, there exists $\delta > 0$ such that for any $x \in (-\delta, \delta)$,

$$\left| \frac{f(x) - f(0)}{x - 0} - f'(0) \right| < \varepsilon.$$

Since $\lim a_n = \lim b_n = 0$, there exists N such that for any $n > N$, $a_n, b_n \in (-\delta, \delta)$. Thus

$$f(a_n) \in (a_n(f'(0) + \varepsilon), a_n(f'(0) - \varepsilon)), \quad f(b_n) \in (b_n(f'(0) - \varepsilon), b_n(f'(0) + \varepsilon)).$$

Hence

$$\begin{aligned} \frac{f(b_n) - f(a_n)}{b_n - a_n} &< \frac{b_n(f'(0) + \varepsilon) - a_n(f'(0) - \varepsilon)}{b_n - a_n} = f'(0) + \frac{b_n + a_n}{b_n - a_n} \varepsilon < f'(0) + \varepsilon, \\ \frac{f(b_n) - f(a_n)}{b_n - a_n} &> \frac{b_n(f'(0) - \varepsilon) - a_n(f'(0) + \varepsilon)}{b_n - a_n} = f'(0) - \frac{b_n + a_n}{b_n - a_n} \varepsilon > f'(0) - \varepsilon. \end{aligned}$$

$$\Rightarrow \left| \frac{f(b_n) - f(a_n)}{b_n - a_n} - f'(0) \right| < \varepsilon.$$

Therefore, $\lim D_n = f'(0)$.

(b) WLOG, assume $f'(0) > 0$. Assume that $b_n < K'(b_n - a_n)$ for some $K' > 1$ and all $n \in \mathbb{N}_+$. Then

$$a_n < \frac{K'}{K' - 1} b_n = K b_n, \text{ where } K = \frac{K'}{K' - 1} \in (0, 1).$$

For any $\varepsilon > 0$, there exists $\delta > 0$ such that for any $x \in (-\delta, \delta)$,

$$\left| \frac{f(x) - f(0)}{x - 0} - f'(0) \right| < \frac{1 - K}{1 + K} \varepsilon.$$

Thus

$$\begin{aligned}
 \frac{f(b_n) - f(a_n)}{b_n - a_n} &= \frac{1}{b_n - a_n} \left(\frac{f(b_n) - f(0)}{b_n - 0} b_n - \frac{f(a_n) - f(0)}{a_n - 0} a_n \right) \\
 &< \frac{1}{b_n - a_n} \left(\left(f'(0) + \frac{1-K}{1+K} \varepsilon \right) b_n - \left(f'(0) - \frac{1-K}{1+K} \varepsilon \right) a_n \right) \\
 &= f'(0) + \frac{1-K}{1+K} \frac{b_n + a_n}{b_n - a_n} \varepsilon \\
 &< f'(0) + \frac{1-K}{1+K} \frac{b_n + Kb_n}{b_n - Kb_n} \varepsilon \\
 &= f'(0) + \varepsilon,
 \end{aligned}$$

and similarly,

$$\frac{f(b_n) - f(a_n)}{b_n - a_n} > f'(0) - \varepsilon.$$

Therefore, $\lim D_n = f'(0)$.

(c) For any $n \in \mathbb{N}_+$, by the mean value theorem, there exists some $\xi_n \in (a_n, b_n)$ such that

$$D_n = \frac{f(b_n) - f(a_n)}{b_n - a_n} = f'(\xi_n).$$

Since $\lim a_n = \lim b_n = 0$, we have $\lim \xi_n = 0$. By the continuity of f' at 0, we have $\lim D_n = \lim f'(\xi_n) = f'(0)$.

Lastly, we give an example which meets the requirement. Take

$$f(x) = x^2 \sin\left(\frac{1}{x}\right),$$

and

$$a_n = \frac{1}{2n\pi + \pi/2}, \quad b_n = \frac{1}{2n\pi}.$$

Then $f'(0) = 0$, and

$$\begin{aligned}
 \frac{f(b_n) - f(a_n)}{b_n - a_n} &= \left(0 - \left(2n\pi + \frac{\pi}{2} \right)^{-2} \right) / \left(\frac{1}{2n\pi} - \frac{1}{2n\pi + \pi/2} \right) \\
 &= - \left(\frac{(2n\pi + \pi/2)^2}{2n\pi} - \left(2n\pi + \frac{\pi}{2} \right) \right)^{-1} \\
 &= - \frac{8n}{(4n+1)\pi} \rightarrow -\frac{2}{\pi} \neq 0. \quad (n \rightarrow \infty)
 \end{aligned}$$

Then $\lim D_n \neq f'(0)$. □

Problem 12 (5-20). State and prove an inequality of the following kind: it can be derived from Taylor's theorem, and it remains valid for vector-valued functions.

Solution. We will prove the following inequality which is easily derived from Taylor's theorem: for any $f : \mathbb{R} \rightarrow \mathbb{R}^n \in C^{k+1}$, $x, y \in \mathbb{R}$ and $k \in \mathbb{N}_+$, we have

$$\left| f(y) - \sum_{i=1}^k \frac{f^{(i)}(x)}{i!} (y-x)^i \right| \leq \frac{1}{(k+1)!} \sup_{0 \leq t \leq 1} |f^{(k+1)}(x + t(y-x))| |y-x|^{k+1}.$$

That is because, from Taylor's theorem with the Lagrange's form of the remainder, there exists some $\xi \in (x, y)$ or (y, x) such that

$$f(y) = \sum_{i=1}^k \frac{f^{(i)}(x)}{i!} (y-x)^i + \frac{f^{(k+1)}(\xi)}{(k+1)!} (y-x)^{k+1}.$$

Combining this with

$$|f^{(k+1)}(\xi)| \leq \sup_{0 \leq t \leq 1} |f^{(k+1)}(x + t(y-x))|,$$

we obtain the desired inequality.

Problem 13 (5-22). Let f be a real function on $(-\infty, \infty)$. If $f(x) = x$, then x is called a fixed point of f .

(a) If f is differentiable and $f'(t) \neq 1$ for every real t , prove that f has at most one fixed point.

(b) Prove that

$$f(t) = t + (1 + e^t)^{-1}$$

has no fixed point, although for every real t one has $0 < f'(t) < 1$.

(c) However, if there is a constant $A < 1$ such that for every real t ,

$$|f'(t)| \leq A,$$

prove that f has a fixed point x , and moreover

$$x = \lim_{n \rightarrow \infty} x_n,$$

where x_1 is any real number and for $n = 1, 2, 3, \dots$,

$$x_{n+1} = f(x_n).$$

(d) Prove that the method in (c) can be implemented along the zig-zag path

$$(x_1, x_2) \rightarrow (x_2, x_2) \rightarrow (x_2, x_3) \rightarrow (x_3, x_3) \rightarrow (x_3, x_4) \rightarrow \dots$$

Proof. (a) Suppose the contrary, if $f(x) = x$ and $f(y) = y$ for some $x \neq y$, then by the mean value theorem, there exists some $c \in (x, y)$ such that

$$f'(c) = \frac{f(y) - f(x)}{y - x} = 1,$$

which contradicts the assumption that $f'(t) \neq 1$ for every real t . Therefore, f has at most one fixed point.

(b) Since $(1 + e^t)^{-1} \neq 0$ for all real t , we have $f(t) = t + (1 + e^t)^{-1} \neq t$ for all real t . Therefore, f has no fixed point.

Note that

$$f'(t) = 1 - \frac{e^t}{(1 + e^t)^2} > 1 - 1 = 0,$$

and obviously $f'(t) < 1$ for all real t . Therefore, $0 < f'(t) < 1$ for all real t .

(c) Suppose the contrary, that is $f(x) \neq x$ for all $x \in \mathbb{R}$. Then by the continuity of f , we have either $f(x) > x$ for all $x \in \mathbb{R}$ or $f(x) < x$ for all $x \in \mathbb{R}$. We only consider the former case, and the latter case can be treated similarly.

If $f(x) > x$ for all $x \in \mathbb{R}$, then for any $x > 0$, there exists some $\xi_x \in (0, x)$, such that

$$f(x) - f(0) = f'(\xi_x)(x - 0) \leq Ax.$$

So we conclude that $x < f(x) \leq Ax + f(0)$ for all $x > 0$, which contradicts the fact that $A < 1$. Therefore, f has a fixed point x .

Next we will prove that $x = \lim_{n \rightarrow \infty} x_n$. For any $n \geq 1$, we have (by the mean value theorem)

$$|x_{n+2} - x_{n+1}| = |f(x_{n+1}) - f(x_n)| = |f'(\xi_n)(x_{n+1} - x_n)| \leq A |x_{n+1} - x_n|.$$

This implies (maybe with a little effort) that $\{x_n\}$ is a Cauchy sequence, and hence converges to some limit x . Then, take $n \rightarrow \infty$ in the equation $x_{n+1} = f(x_n)$, we get $x = f(x)$, which implies that x is the fixed point of f . Therefore, $x = \lim_{n \rightarrow \infty} x_n$.

(d) Actually, the method in (c) is exactly the same as the zig-zag path described in (d). Therefore, the conclusion in (c) also holds for the zig-zag path. $\square$

Problem 14 (5-23). Let

$$f(x) = \frac{x^3 + 1}{3}.$$

This function has three fixed points, say α, β, γ , with

$$-2 < \alpha < -1, \quad 0 < \beta < 1, \quad 1 < \gamma < 2.$$

Pick any x_1 and define $\{x_n\}$ by $x_{n+1} = f(x_n)$.

(a) If $x_1 < \alpha$, prove that $x_n \rightarrow -\infty$ as $n \rightarrow \infty$.

(b) If $\alpha < x_1 < \gamma$, prove that $x_n \rightarrow \beta$ as $n \rightarrow \infty$.

(c) If $\gamma < x_1$, prove that $x_n \rightarrow +\infty$ as $n \rightarrow \infty$.

Hence β can be found by this method, but α and γ cannot.

Proof. (a) We will prove the following inequality first: for any $x < \alpha$, we have $f(x) < x < \alpha$ and

$$|f(x) - \alpha| > f'(\alpha) |x - \alpha|.$$

The first inequality is the direct consequence of the fact that α is the least fixed point of f and $f(x)$ is increasing on $(-\infty, \infty)$. For the second inequality, note that for any $x < \alpha$,

$$f'(x) = x^2 > \alpha^2 = f'(\alpha).$$

Then by the mean value theorem, there exists some $\xi_x \in (x, \alpha)$ such that

$$|f(x) - \alpha| = |f(x) - f(\alpha)| = |f'(\xi_x)(x - \alpha)| > |f'(\alpha)(x - \alpha)|.$$

Combining this inequality with $f'(\alpha) = \alpha^2 > 1$, we get

$$|x_n - \alpha| > (f'(\alpha))^{n-1} |x_1 - \alpha| \rightarrow +\infty \implies x_n \rightarrow -\infty.$$

(c) The proof is rather similar to (a). We only need to prove the following inequality: for any $x > \gamma$, we have $f(x) > x > \gamma$ and

$$|f(x) - \gamma| > f'(\gamma) |x - \gamma|.$$

Then the conclusion follows.

(b) We will first deal with the case where $x_1 \in (\alpha, \beta)$. Since $f(x)$ is upper-convex on $(-\infty, -1)$, we have

$$f(x) > \frac{f(-1) - f(\alpha)}{-1 - \alpha}(x - \alpha) + f(\alpha) \implies f(x) - \alpha > k_1(x - \alpha), \text{ where } k_1 = \frac{f(-1) - f(\alpha)}{-1 - \alpha} > 1.$$

So, if $x_1 \in (\alpha, -1)$, then for any $k \geq 1$ such that $x_k \in (\alpha, -1)$, we have

$$x_{k+1} - \alpha > k(x_k - \alpha) \implies x_k > k_1^{k-1}(x_1 - \alpha) + \alpha.$$

This implies that there exists some N such that $x_N \in (-1, \beta)$. So we can assume that $x_1 \in (-1, \beta)$.

Then, we will prove the following inequality: for any $x \in (-1, \beta)$, we have

$$\beta - f(x) < k_2(\beta - x), \text{ where } k_2 = \frac{f(\beta) - f(-1)}{\beta + 1} = \frac{\beta}{\beta + 1} < 1.$$

Note that this inequality is equivalent to

$$f(x) > k_2(x + 1),$$

so it suffices to prove the inequality above.

- If $x \in (-1, 0)$, since f is upper-convex on $(-\infty, -1)$, we have

$$f(x) > \frac{f(0) - f(-1)}{0 - (-1)}(x - (-1)) = \frac{1}{3}(x + 1) > k_2(x + 1).$$

- If $x \in (0, \beta)$, we have $f'(x) = x^2 < \beta^2 < \beta/(\beta + 1) = k_2$, so by the mean value theorem, there exists some

$\xi_x \in (x, \beta)$ such that

$$f(x) = f(\beta) - f'(\xi_x)(\beta - x) > \beta - k_2(\beta - x) \implies \beta - f(x) < k_2(\beta - x).$$

Then, for any $k \geq 1$, we have

$$\beta - x_{N+k+1} < k_2(\beta - x_{N+k}) \implies \beta - x_{N+k} < k_2^k(\beta - x_N) \rightarrow 0. \quad (k \rightarrow \infty)$$

Therefore, $x_n \rightarrow \beta$.

Next, we will deal with the case where $x_1 \in (\beta, \gamma)$. It's clear that $x_1 > x_2 > \dots > \beta$. Then, since f is lower-convex on (β, ∞) , for any $x \in (\beta, x_1)$, we have

$$f(x) < k_3(x - \beta) + \beta, \text{ where } k_3 = \frac{f(x_1) - \beta}{x_1 - \beta} < 1.$$

So, for any $k \geq 1$, we have

$$x_{k+1} - \beta < f(x_k) - \beta < k_3(x_k - \beta) \implies x_k - \beta < k_3^{k-1}(x_1 - \beta) \rightarrow 0. \quad (k \rightarrow \infty)$$

This implies that $x_n \rightarrow \beta$. □

Problem 15 (5-25). Let f be twice differentiable on $[a, b]$, with $f(a) < 0$, $f(b) > 0$. Assume that for all $x \in [a, b]$,

$$f'(x) \geq \delta > 0 \quad \text{and} \quad 0 \leq f''(x) \leq M.$$

Let ξ be the unique point in (a, b) such that $f(\xi) = 0$. Consider the Newton iteration and complete the details below.

(a) Choose $x_1 \in (\xi, b)$ and define

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

Use a geometric interpretation via the tangent line of the graph of f .

(b) Prove that $x_{n+1} < x_n$ and $\lim_{n \rightarrow \infty} x_n = \xi$.

(c) Use Taylor's theorem to show that there exists $t_n \in (\xi, x_n)$ such that

$$x_{n+1} - \xi = \frac{f''(t_n)}{2f'(x_n)}(x_n - \xi)^2.$$

(d) Let $A = M/(2\delta)$ and derive the inequality

$$0 \leq x_{n+1} - \xi \leq \frac{1}{A}[A(x_1 - \xi)]^{2^n}.$$

Compare with Chapter 3, Exercises 16 and 18.

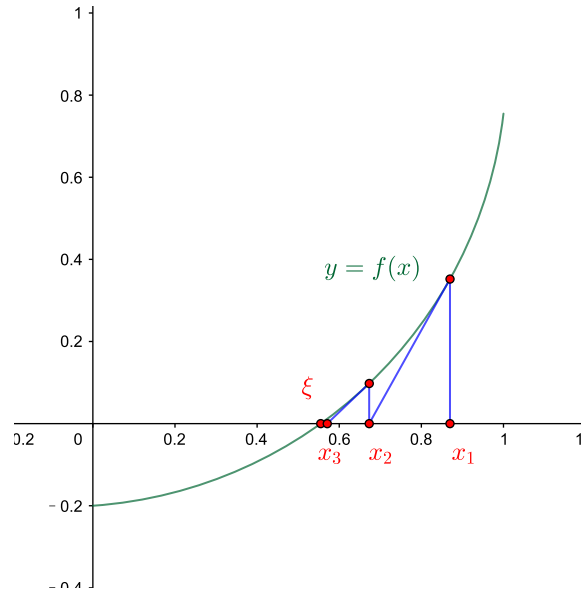
(e) Prove that Newton's method is equivalent to finding fixed points of

$$g(x) = x - \frac{f(x)}{f'(x)}.$$

What is the behavior of $g'(x)$ as x approaches ξ ?

(f) Let $f(x) = x^{1/3}$ for $x \in (-\infty, \infty)$. What happens to Newton's method?

Proof. (a) As shown below.



(b) First, we will show that for any $x_0 \in (\xi, b)$, we have

$$x_0 - \frac{f(x_0)}{f'(x_0)} \geq \xi.$$

Note that $f'(x)$ is increasing on (a, b) , since $f''(x) \geq 0$. If $x_1 := x_0 - (f(x_0))/(f'(x_0)) < \xi$, then by the mean value theorem, there exists some $\eta \in (\xi, x_0)$ such that

$$f'(\eta) = \frac{f(x_0) - f(\xi)}{x_0 - \xi} > \frac{f(x_0)}{x_0 - x_1} = f'(x_0),$$

which contradicts the fact that $\eta < x_0$. Therefore, $x_1 \geq \xi$. Then, by induction, we can prove that $x_n \geq \xi$ for all $n \in \mathbb{N}_+$. Moreover, since f is monotone increasing on (ξ, b) , we have $f(x_n) \geq 0$ for all $n \in \mathbb{N}_+$. Hence,

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} < x_n.$$

Since $\{x_n\}$ is decreasing and bounded, it converges to some limit x . Then, take $n \rightarrow \infty$ in the equation $x_{n+1} = x_n - (f(x_n))/(f'(x_n))$, we get $x = x - (f(x))/(f'(x))$, which implies that $f(x) = 0$. Since ξ is the unique point in (a, b) such that $f(\xi) = 0$, we have $x = \xi$. Therefore, $\lim_{n \rightarrow \infty} x_n = \xi$.

(c) Use Taylor's theorem with the Lagrange's form of the remainder at x_n , there exists some $t_n \in (\xi, x_n)$ such that

$$f(\xi) - (f(x_n) + f'(x_n)(\xi - x_n)) = \frac{f''(t_n)}{2}(\xi - x_n)^2.$$

By the definition of x_{n+1} , we have

$$\begin{aligned} f(\xi) - (f(x_n) + f'(x_n)(\xi - x_n)) &= f'(x_n) \left(x_n - \xi - \frac{f(x_n)}{f'(x_n)} \right) \\ &= f'(x_n)(x_{n+1} - \xi). \end{aligned}$$

Therefore

$$x_{n+1} - \xi = \frac{f''(t_n)}{2f'(x_n)}(\xi - x_n)^2.$$

(d) Since $f''(x_n) \leq M$ and $f'(x_n) \geq \delta$, we have

$$x_{n+1} - \xi \leq \frac{M}{2\delta}(\xi - x_n)^2 \implies (A(x_{n+1} - \xi)) \leq (A(x_n - \xi))^2.$$

By induction, we get

$$0 \leq A(x_{n+1} - \xi) \leq [A(x_1 - \xi)]^{2^n}.$$

(e) Take a fixed point of $g(x)$, say ξ . Then

$$0 = \xi - g(\xi) = \frac{f(\xi)}{f'(\xi)} \implies f(\xi) = 0.$$

Note that $g(x) = x - (f(x))/(f'(x))$, so $g'(x) = 1 - (f'(x)f'(x) - f(x)f''(x))/(f'(x))^2$. Since $f(\xi) = 0$, we have $g'(\xi) = 0$. Therefore, as x approaches ξ , $g'(x)$ approaches 0.

(f) Newton's method may be invalid for this function, since $f'(0)$ does not exist. □